SPH4U Unit 3 Gravitational, Electric, and Magnetic Field

Chapter 8 Magnetic Fields Lesson Package

Schoolwork	Useful Links/ Videos/ Handouts	Homework
8.1 Magnets and Electromagnets		
1. Lesson 8.1 Presentation 2. Lesson 8.1 Lesson Package	**Videos:** 1. 8.1 Magnets and electromagnets 2. Magnetic Fields – Observing Magnetic Force	1. Textbook pg. 385 #1-5
8.2 Magnetic Force on Moving Charges		
1. Lesson 8.2 Presentation 2. Lesson 8.2 Lesson Package	**Videos:** 1. 8.2 Magnetic force on moving charges	1. Textbook pg. 391 #1-10
8.3 Magnetic Force on a Current-Carrying Conductor		
1. Lesson 8.3 Presentation 2. Lesson 8.3 Lesson Package	**Videos:** 1. 8.3 Magnetic force on a current-carrying conductor	1. Textbook pg. 396 #1-5
8.4 Motion of Charged Particles in Magnetic Fields		
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8.5 Applications of Electric and Magnetic Fields		
	Videos: 1.	1. Lesson 8.5 Presentation 2. read the Textbook pg. 405-409

Chapter 8 Review		
1. Chapter 8 Self- Quiz pg. 415 (attempt all) 2. Chapter 8 Review pg. 416 #1-65 3. Communication Questions (lesson package)		1. Chapter 8 Self- Quiz pg. 415 (attempt all) 2. Chapter 8 Review pg. 416 #1-65 3. Chapter 8 - Check Your Knowledge – Definitions (lesson package) 4. Communication Questions (lesson package)

Lesson 8.1 Magnets and Electromagnets

Lesson Goal: ____

An **electromagnet** any device that produces a magnetic field as a result of an electric current

The benefit of the electromagnet is that it can be switched on and off and the strength can be increased.

The strength of the electromagnet can be increased by:

- increasing the number of loops in the coil,

- increasing the current, or

- introducing a core made from a material that is quickly magnetized. Soft iron is such a material, and it can be just as quickly de-magnetized when the current is switched off. A core material like soft iron concentrates the magnetic field.

To make a very powerful magnet, we include all three factors. The most powerful electromagnets have several thousand loops of wire, work with large currents, and have a soft-iron core.

Electromagnets have many applications:

• They are used in scrap metal yards to pick up and drop large metal objects such as cars.

• They are used in washing machines and dishwashers to regulate the flow of water.

• They are used in doorbells to pull a lever against a bell and release it.

• Electromagnets also form the central piece of the MRI unit you read about at the start of this chapter.

Video:

1. Magnetic Field Demonstration

2. Iron Filings Demonstration

8.2 Magnetic Force on Moving Charges

Lesson Goal: ____

The unit of magnetic field strength in the SI system is the tesla (T). A tesla is defined in terms of other SI units by this conversion:

1T = 1 kg/(C*s)

A Charge in a Magnetic Field

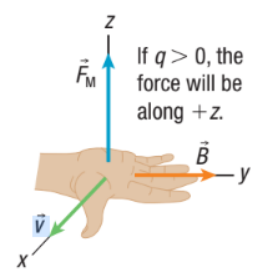
Magnetic forces act on all types of electric charges, such as electrons, protons, and ions. An important property of the magnetic force is that it depends on the velocity of the charge. The magnetic force is non-zero only if a charge is in motion. This property of the magnetic force is not shared by the gravitational force and the electric force. Those forces are both independent of velocity.

$F_M = qvB\sin\theta$

The magnetic force, just like any other force, is a vector quantity. The above equation provides only the magnitude of the magnetic force. We can determine the direction of the force using the right-hand rule for a moving charge in a magnetic field.

Right-Hand Rule for a Moving Charge in a Magnetic Field

If you point your right thumb in the direction of the velocity of the charge ($\vec{v}$), and your straight fingers in the direction of the magnetic field ($\vec{B}$), then your palm will point in the direction of the resulting magnetic force ($\vec{F}_M$).



Characteristics of the magnetic force:

- Depends on velocity.
- Unlike gravitational and electric forces, the direction of F_M is always perpendicular to both the magnetic field and the particle's velocity. The direction in which the force acts and the direction of the field are always parallel for electric and gravitational fields.
- Since F_M is always perpendicular to $\vec{v}$, it is a purely deflecting force meaning it changes the direction of $\vec{v}$ but has no effect on the magnitude of the velocity or speed. This is true because no component of the force acts in the direction of motion of the charged particle.

Video: [Magnetic Force on Current Carrying Wire]

Example 1: Magnetic Force on a Negative Charge in Motion

The electron in figure below moves at a speed of 54 m/s through a magnetic field with a strength of 1.2 T. The angle between the electron's velocity vector and the magnetic field is 90°. Assume a value for the electron's charge of $q = -e = -1.60 * 10^{-19}$ C.

a) What is the magnitude of the magnetic force on the electron? Express your answer in newtons (N), using 1 T = 1

Kg/(C*s)

b) Use the right-hand rule to determine the direction of the magnetic force.

c) Calculate the gravitational force on the electron. The mass of an electron is $9.11 * 10^{-31}$ kg.

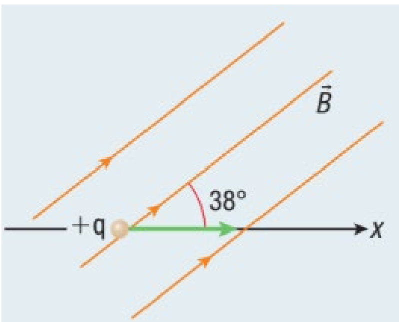
d) What is the ratio of the gravitational force on the electron to the magnetic force on the electron?

Example 2: Magnetic Force on a Positive Charge in Motion

A proton is moving along the x-axis at a speed of 78 m/s. It enters a magnetic field of strength 2.7 T. The angle between the proton's velocity vector and the magnetic field is 38°. The mass of a proton is $1.67 * 10^{-27}$ kg.

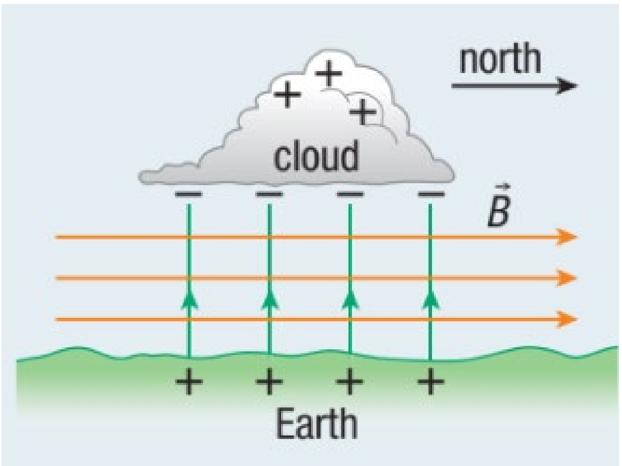
a) Calculate the initial magnitude and the direction of the magnetic force on the proton.

b) Determine the proton's initial acceleration.



Example 3: Determining the Effect of Earth's Magnetic Field on the Direction of Lightning Strikes

During a thunderstorm, positive charge accumulates near the top of a cloud, and negative charge accumulates near the bottom of a cloud (figure below). When the charge buildup is strong enough, negative charge moves rapidly from the cloud to the ground as a lightning strike. Assume the charge velocity vector is perpendicular to the ground, and Earth's magnetic field is horizontal, directed north. Determine the direction of the deflection of this charge by Earth's magnetic field.



8.3 Magnetic Force on a Current-Carrying Conductor

Lesson Goal: ____

Figure 2 This current-carrying wire is in an external magnetic field. The magnetic field is perpendicular to the wire and is directed out of the page as indicated by the dots.

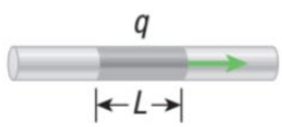
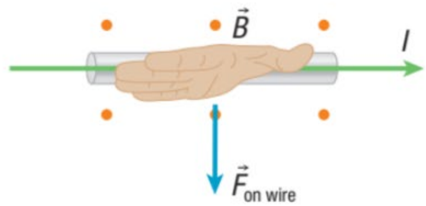


Figure 3 In our analysis to determine the magnetic force on a current-carrying wire, we examine a section of the wire and call it length L.

Use the right-hand rule to determine the direction of $\vec{F}_{\text{on wire}}$.

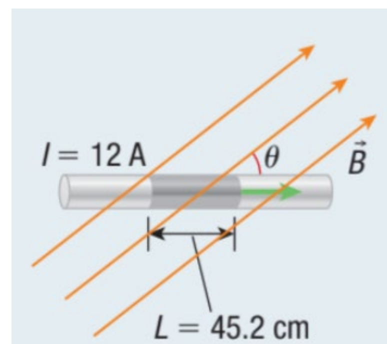
The magnetic force on a current is due to the force on a moving charge, so the right-hand rule used here is similar to other right-hand rules you have used before. The direction of the current replaces the direction of $\vec{v}$ for a moving charge.



You can use the right-hand rule to determine the direction of the magnetic force on the wire.

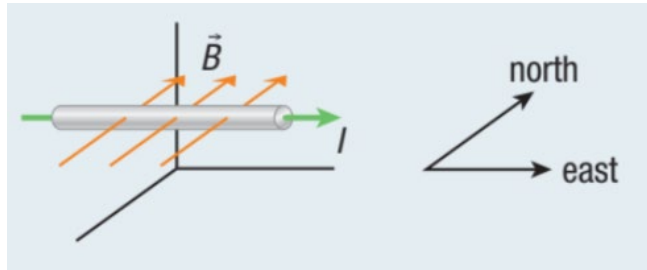
Example 1: Calculating the Magnitude of the Magnetic Force on a Wire in a Uniform Magnetic Field

A piece of wire 45.2 cm long has a current of 12 A. The wire moves through a uniform magnetic field with a strength of 0.30 T. Calculate the magnitude of the magnetic force on the wire when the angle between the magnetic field and the wire is (a) 0°, (b) 45°, and (c) 90°.



Example 2: Determining Magnetic Force on a Segment of a Current-Carrying Wire in Earth's Magnetic Field

Two electrical poles support a current-carrying wire. The mass of a 2.5 m segment of the wire is 0.44 kg. A 15 A current travels through the wire. The conventional current is oriented due east, horizontal to Earth's surface. The strength of Earth's magnetic field at the location is 57 µT and is oriented due north, horizontal to Earth's surface.



a) Determine the magnitude and the direction of the magnetic force on the 2.5 m segment of wire.

b) Calculate the gravitational force on the 2.5 m segment of wire.

8.4 Motion of Charged Particles in Magnetic Fields

Lesson Goal: ____

Charges and Uniform Circular Motion

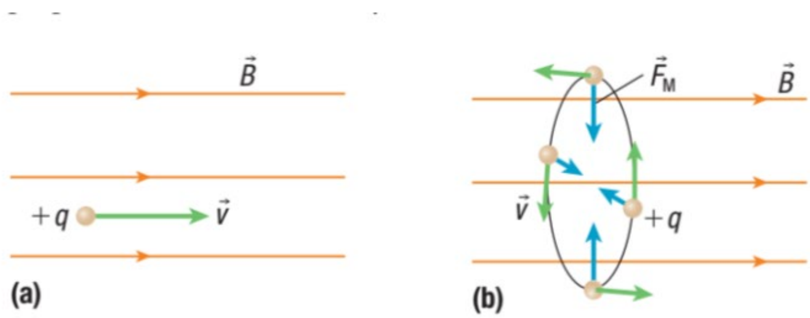


Figure 2 (a) When the velocity of a charged particle is parallel to the magnetic field $\vec{B}$, the magnetic force on the particle is zero. (b) When $\vec{v}$ makes a right angle with $\vec{B}$ ($\theta = 90°$), the charged particle moves in a circle that lies in a plane perpendicular to $\vec{B}$.

Example 1: An Electron in a Magnetic Field

An electron starts from rest. A horizontally directed electric field accelerates the electron through a potential difference of 37 V. The electron then leaves the electric field and moves into a magnetic field. The magnetic field strength is 0.26 T, directed into the page, and the mass of the electron is $9.11 * 10^{-31}$ kg.

a) Determine the speed of the electron at the moment it enters the magnetic field.

b) Determine the magnitude and direction of the magnetic force on the electron.

c) Determine the radius of the electron's circular path.

Example 2: The Mass Spectrometer: Identifying Particles

A researcher using a mass spectrometer observes a particle travelling at $1.6 * 10^6$ m/s in a circular path of radius 8.2 cm. The spectrometer's magnetic field is perpendicular to the particle's path and has a magnitude of 0.41 T.

a) Calculate the mass-to-charge ratio of the particle. (In 1910, Robert Millikan accurately determined the charge carried by an electron. His finding allowed researchers to calculate the mass of charged particles using the mass-to-charge ratio.)

b) Identify the particle using Table 1.

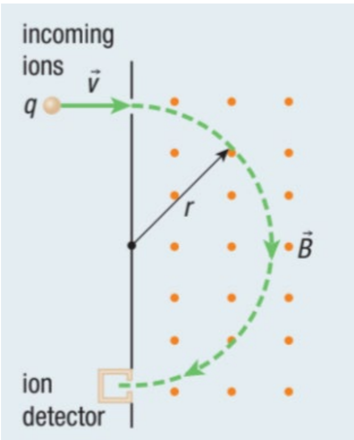
Table 1

Isotope	m (kg)	q (C)	$\frac{m}{q}$ (kg/C)
hydrogen	1.67×10^{-27}	1.60×10^{-19}	1.04×10^{-8}
deuterium	3.35×10^{-27}	1.60×10^{-19}	2.09×10^{-8}
tritium	5.01×10^{-27}	1.60×10^{-19}	3.13×10^{-8}

Example 3: The Mass Spectrometer: Separating Isotopes

A researcher uses a mass spectrometer in a carbon dating experiment (figure below). The incoming ions are a mixture of $^{12}C^{+}$ and $^{14}C^{+}$, and they have speed v = 1.0 * 10^5 m/s. The strength of the magnetic field is 0.10 T. The mass of the electron is 9.11 * 10^{-31} kg. The mass of the proton and the mass of the neutron are both 1.67 * 10^{-27} kg.

The researcher first positions the ion detector to determine the value of r for $^{12}C^{+}$ and then moves it to determine the value of r for $^{14}C^{+}$. How far must the detector move between detecting $^{12}C^{+}$ and $^{14}C^{+}$?



Field Theory

Field theory a scientific model that describes forces in terms of entities that exist at every point in space.

Studying gravitational, electric, and magnetic forces has revealed differences and similarities between these forces and their respective fields.

Chapter 8 - Check Your Knowledge – Definitions

Term	Definition
___ field theory	1. If your thumb is pointing in the direction of conventional current, and you curl your fingers forward, your curled fingers point in the direction of the magnetic field lines.
____ principle of electromagnetism	2. a technology that uses microchips that act as transmitters and responders to communicate data by radio waves
____ magnetorheological fluid	3. the SI unit of measure for describing the strength of a magnetic field; equivalent to 1 kg/C*s
____ right-hand rule for a solenoid	4. one of a set of lines drawn to indicate the strength and direction of a magnetic field
____ magnetic field line	5. A process in which magnetic fields interact with atoms in the human body, producing images that doctors can use to diagnose injuries and diseases
____ right-hand rule for a moving charge in a magnetic field	6. If you point your thumb in the direction of the velocity of the charge, and your straight fingers in the direction of the magnetic field, then your palm will point in the direction if the resulting magnetic force.
____ radio-frequency identification technology (RFID)	7. a scientific model that describes forces in terms of entities that exist at every point in space
____ right-hand rule for a straight conductor	8. If you coil the fingers of your right hand around a solenoid in the direction of the conventional current, your thumb points in the direction of the magnetic field lines in the centre of the coil.
___ tesla	9. moving electric charges produce a magnetic field
____ magnetic resonance imaging (MRI)	10. a fluid containing suspended iron particles that, when subjected to a magnetic field, changes to solid

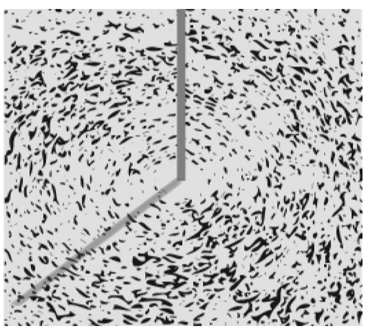
Chapter 8 Communication Questions

1. Why would an electromagnet be used in a scrap-metal recycling operation rather than a permanent magnet? Suggest at least three reasons. (OBJ: 8.1 Magnets and Electromagnets)

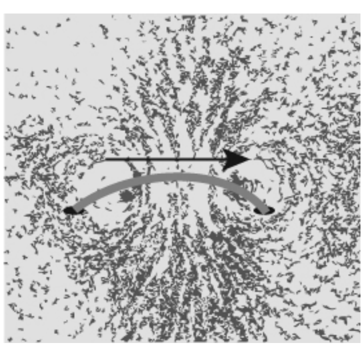
2. Suggest a reason why the aurora borealis moves and swirls, instead of remaining fixed. (OBJ: 8.1 Magnets and Electromagnets)

3. The magnetic field of Earth can be considered as an example of electromagnetism. Explain why this is so. (OBJ: 8.1 Magnets and Electromagnets)

4. A hole is punched in a piece of paper and a wire is placed in the hole, perpendicular to the paper. A current flows through the wire, and forms magnetic field lines that run clockwise around the wire, as shown. Which way is the current flowing in the wire? Explain your answer. (OBJ: 8.1 Magnets and Electromagnets)



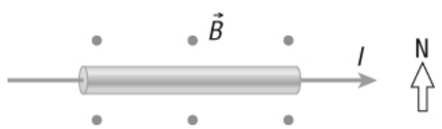
5. A loop of copper is placed through two holes in a piece of paper. A current through the loop as shown forms a magnetic field. Which way do the field lines pass through the centre of the loop? Explain your answer. (OBJ: 8.1 Magnets and Electromagnets)



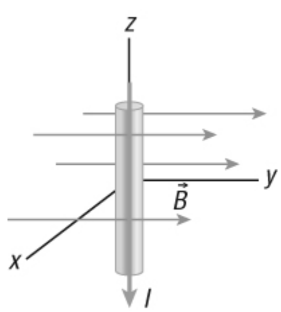
6. A proton and an electron are shot into a magnetic field perpendicular to the field lines at the same speed. How does the magnetic force on the proton compare to the magnetic force on the electron?
(OBJ: 8.2 Magnetic Force on Moving Charges)

7. A proton and an alpha particle are shot into a magnetic field at the same velocity relative to the field. The alpha particle has twice the charge and four times the mass of the proton. How does the magnetic force on the alpha particle compare to the magnetic force on the proton? (OBJ: 8.2 Magnetic Force on Moving Charges)

8. A current flows in a wire from west to east, as shown. A magnetic field points upward out of the page. In what direction is the magnetic force acting on the wire? (OBJ: 8.3 Magnetic Force on a Current-Carrying Conductor)



9. A segment of wire carries a current toward the negative *z*-axis, as shown. A magnetic field point in the direction of the positive *y*-axis. In what direction is the force on the wire? Explain your answer.
(OBJ: 8.3 Magnetic Force on a Current-Carrying Conductor)



10. Explain why a charged particle moving in a magnetic field with a velocity that is neither parallel nor perpendicular to the field should move in a spiral path. Include a discussion of vector components in your answer.
(OBJ: 8.4 Motion of Charged Particles in Magnetic Fields)

11. The Van Allen belts contain charged particles that are trapped in Earth's magnetic field. Most of the trapped particles are located over the equator. Suggest a reason why this might be so.
(OBJ: 8.4 Motion of Charged Particles in Magnetic Fields)

12. Large massive objects like planets or stars are affected mostly by gravitational forces, while small objects like electrons and protons are affected more by electric or magnetic forces. Suggest reasons why this is so.
(OBJ: 8.4 Motion of Charged Particles in Magnetic Fields)

13. During the development of the atomic bomb in World War 2, scientists tried to use a mass spectrometer to separate the isotopes of uranium in order to concentrate the isotope they needed. Use your knowledge of isotopes, and explain why this might work. (OBJ: 8.4 Motion of Charged Particles in Magnetic Fields)

14. Explain how the operation of a RFID chip is different from the magnetic stripe currently used on many credit cards or driver licences. (OBJ: 8.5 Applications of Electric and Magnetic Fields)

15. Explain how an MR fluid changes from a liquid to a solid. (OBJ: 8.5 Applications of Electric and Magnetic Fields)

16. Why does a correlation between two measured quantities not prove cause and effect? Give an example to support your answer. (OBJ: 8.5 Applications of Electric and Magnetic Fields)

17. Give an example of a product that cannot use an embedded RFID chip for data storage. Explain why this is so. (OBJ: 8.5 Applications of Electric and Magnetic Fields)

18. Explain why an MRI machine works well with tissues from a living human body, but would not work well with a mummy. (OBJ: 8.5 Applications of Electric and Magnetic Fields)

19. How does an MRI machine help medical personnel to detect abnormalities such as cancerous tumours? (OBJ: 8.5 Applications of Electric and Magnetic Fields)

20. A friend who has not studied physics challenges you to provide evidence that magnetic fields exist around an electromagnet. Describe a demonstration that you could perform on the spot to illustrate the existence of such a field. (OBJ: 8.1 Magnets and Electromagnets)